🦕 Famous Fossils & Deep Time: Reading · ANSWER KEY

Biology A · Unit 7 · Level 3 (closer to the class work) · goes with Unit 4 Famous Fossils: A4.1 WA fossil stations, A4.2 Twine Time, A4.3 Building Bridges, rock layers game

Name: _____

Reading the Story in the Rocks

A fossil is the remains or trace of a living thing from long ago, kept in rock. For a fossil to form, the organism must be buried quickly in fine sediment like mud, sand or volcanic ash. Over a long time, minerals replace hard parts like bone, shell and wood. Some fossils are molds, imprints or footprints instead of the body itself.

Rock forms in layers called strata. Older layers are deeper and younger layers are on top. When scientists chart where a species appears, it shows up as a column through the layers. When the species goes extinct, its column ends.

Washington has famous fossils. The Columbian mammoth is the state fossil, chosen with help from elementary students. The first Washington dinosaur bone, from a theropod, was found in the San Juan Islands in 2012. Clovis stone tools found in East Wenatchee are about 11,000 to 12,000 years old. The Blue Lake Rhino is a mold in lava rock, and the Ginkgo Petrified Forest near Vantage has wood turned to stone.

Earth is about 4.6 billion years old, and life began about 3.5 billion years ago. In the Twine Time activity, a long piece of twine stands for all of Earth's history. Humans appear only at the very end, a tiny piece of the twine. This huge span is called deep time.

For millions of years North and South America were separate. Plate movement built the Central American land bridge, and animals crossed it: armadillos and opossums moved north. Paleontologists, scientists who study fossils, use these clues to learn how life changed.

Check your reading

1. What has to happen for a fossil to form?
The organism is buried quickly in fine sediment, and minerals slowly replace hard parts.
2. In rock layers, where are the oldest fossils?
In the deepest (bottom) layers.
3. What is Washington's state fossil?
The Columbian mammoth.